A Reworking of Mochizuki's Hom Theorem over p-Subtame Fields

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Abstract

We rework Mochizuki's proof of the anabelian Hom theorem, following also Faltings's exposition of its Picard-theoretic arguments. Let p be a prime, and let K be a subfield of the maximal tamely ramified extension of a henselian discretely valued field of mixed characteristic $(0, p)$. For every set of primes Σ containing p, we prove that dominant morphisms from a normal geometrically integral K-scheme of finite type to a hyperbolic curve are determined by, and exhaust, the open homomorphisms of their geometrically pro-Σ arithmetic fundamental groups over the absolute Galois group of K, up to geometric inner automorphisms. The local argument uses Tate's theorem on p-divisible groups and the relative Picard scheme of a semistable curve. Approximation of Kummer classes then permits the reconstruction of a morphism from its action on regular differentials.

1 Introduction

Grothendieck's anabelian programme asks whether arithmetic fundamental groups recover varieties and their morphisms [6]. For hyperbolic curves, the early rigidity results of Nakamura for punctured projective lines [14] were followed by Tamagawa's Isom theorem for affine curves over fields finitely generated over $\mathbf{Q}$ [17, Theorem (0.3)] and Mochizuki's Isom theorem for proper curves over number fields [11, Theorem A]. Mochizuki subsequently proved the local pro-p Hom theorem, which recovers dominant morphisms to hyperbolic curves from open homomorphisms of arithmetic fundamental groups. Its ground field is sub-p-adic, that is, a subfield of a finitely generated extension of $\mathbf{Q}_p$, and its source is smooth [12, Theorem A and Theorem 16.5]. The theorem remains true after replacing the geometric fundamental groups by their maximal pro-Σ quotients, for any set of primes Σ containing p; this formulation is stated in [13, Theorem A and the subsequent remark, p. 121].

The present paper revisits this proof, with particular attention to the hypothesis on the ground field. The reconstruction through Jacobians, Kummer classes, and regular differentials is Mochizuki's. Our presentation also follows Faltings's account of the Picard-torsor and point-approximation arguments [4, §§5–6]. The use of points on successive covers has its antecedent in Tamagawa's characterization of geometric sections [17, Corollary (2.10)]; its adaptation to the local Hom theorem is developed in [12, §§7–10]. We carry out the local comparison over complete discrete valuation rings and their tamely ramified extensions. This leads to the following class of fields.

Definition 1.1. Fix a prime p. A field K of characteristic zero is *p-subtame* if there exist a henselian discretely valued field F of mixed characteristic $(0, p)$ and an embedding

$$K \hookrightarrow F^{\mathrm{t}},$$

where F^{t} denotes its maximal tamely ramified algebraic extension. The extension F^{t}/F is algebraic; its completion is not used in this definition. The residue field of F is arbitrary.

One obtains the same class of fields by requiring F to be complete. Moreover, the class is closed under subfields and finitely generated extensions (Lemma 2.1). In particular it contains the sub-p-adic fields and the generalized sub-p-adic fields of [13, Definition 4.11]. Mochizuki's theorem over the latter class is an Isom theorem [13, Theorem 4.12]. We prove the Hom statement over p-subtame fields by the local argument developed below.

Fix an algebraic closure $\bar{K}$ of K, and put $G_K = \mathrm{Gal}(\bar{K}/K)$. For a nonempty set of primes Σ and a geometrically connected K-scheme Z, write

$$\Delta_Z^{\Sigma} = \pi_1(Z_{\bar{K}})^{(\Sigma)}, \qquad \Pi_Z^{\Sigma} = \pi_1(Z) \big/ \ker\big(\pi_1(Z_{\bar{K}}) \longrightarrow \Delta_Z^{\Sigma}\big).$$

Here (Σ) means the maximal pro-Σ quotient. Thus

$$1 \longrightarrow \Delta_Z^{\Sigma} \longrightarrow \Pi_Z^{\Sigma} \longrightarrow G_K \longrightarrow 1. \tag{1.1}$$

All homomorphisms of profinite groups are understood to be continuous. An open homomorphism is one with open image. We suppress geometric base points; the quotient by geometric inner automorphisms in the theorem accounts for their choice. The subscript G_K on Hom means that the homomorphisms commute with the projections to G_K. A hyperbolic curve is a smooth separated geometrically connected curve whose smooth proper compactification has genus g and whose boundary has r geometric points, with $2g - 2 + r > 0$.

Theorem 1.2 (Geometrically pro-Σ Hom theorem)**.** *Let K be p-subtame, and let Σ be any set of primes containing p. Let X be a normal geometrically integral finite-type K-scheme and Y a hyperbolic curve over K. Then the natural map*

$$\mathrm{Hom}_K^{\mathrm{dom}}(X, Y) \;\xrightarrow{\sim}\; \mathrm{Hom}_{G_K}^{\mathrm{open}}(\Pi_X^{\Sigma}, \Pi_Y^{\Sigma}) / \mathrm{Inn}(\Delta_Y^{\Sigma}) \tag{1.2}$$

is bijective. The scheme X need not be separated.

For $\Sigma = \{p\}$ this is the geometrically pro-p Hom theorem. Taking Σ to be the set of all primes gives the corresponding statement for full étalefundamental groups. In both cases the arithmetic quotient in (1.1) is the full group G_K.

Other extensions of anabelian Isom results use different hypotheses on the ground field. Hoshi proves an Isom theorem for affine hyperbolic curves over Kummer-faithful fields under point-theoreticity and Galois-preservation conditions [8, Theorem A]. Tsujimura proves a relative birational Isom theorem for curves over arbitrary cyclotomic extensions of number fields [19, Corollary B]. A relative birational Hom analogue over generalized sub-p-adic fields is posed as a question in [20, Introduction, Remark (a)]. Theorem 1.2 concerns arithmetic fundamental groups of finite-type schemes, with their projections to the fixed group G_K. Mochizuki's formulation for pro-varieties also includes function fields [12, Definition 16.4 and Theorem 16.5].

The change in the ground field requires a local argument valid without any perfectness assumption on the residue field. We prove the needed comparison using connected p-divisible groups, finite Kummer torsors, and an integral estimate on differentials. The reconstruction is then expressed in terms of approximation at each finite level. We also give an extension criterion for normal sources, including singular ones. The finite-cover method and the passage from pro-p to pro-Σ remain those of the preceding works; the latter is already part of Mochizuki's formulation cited above.

Outline of the proof

Consider first a surjective homomorphism between the geometrically pro-p arithmetic fundamental groups of two smooth proper curves, with hyperbolic target. Abelianization gives

a homomorphism between the Tate modules of their Jacobians. To recover a morphism of curves, one needs first to turn this homomorphism into a map on regular differentials, and then to show that the map on differentials respects the equations of the target curve. This is the order of reconstruction in Mochizuki's proof [12, §§7–14].

The first question is local. After embedding K in a tame extension of a complete discretely valued field, the Tate-module homomorphism descends to a finite extension of that field. The reason is that a prime-to-p group acting on a $\mathbf{Z}_p$-lattice is detected by reduction modulo p (Lemma 2.2). On a regular semistable model, inertia characterizes the Tate module of the connected formal subgroup of the Jacobian. Tate's full faithfulness theorem therefore gives an integral homomorphism of formal groups, whose differential is the required map [18, §4.2, Theorem 4 and Corollary 1]. Section 4 also proves that this homomorphism preserves finite Kummer classes. The point here is to subtract a point defined over the algebraic constant field before passing to the formal group. This subtraction gives both the assertion about Kummer classes and an integral bound on the error in the differential comparison.

One then applies the Kummer assertion on finite covers of the target. Over a suitable field with pro-p absolute Galois group, the resulting sections can be approximated, modulo any prescribed power of p, by sections arising from points (Section 5). Choose a generic point of the source in a Gauss-valued extension and a derivation which preserves its valuation ring. The differential values at the approximating points converge to the values prescribed by the Tate-module homomorphism. Ratios of canonical differentials, together with their first derivatives, generate an affine chart of the target; the derivatives supply the functions missing from the canonical map in the hyperelliptic case. Passing to the limit in their polynomial relations produces a morphism of curves (Proposition 6.2). Thus only approximation at each finite level is needed.

This construction initially takes place over an algebraically closed extension of K. Applying it to finite covers identifies the geometric outer homomorphism induced by the constructed morphism. Faithfulness of the action on differentials then gives descent to K, and the triviality of centralizers of open subgroups in the geometric target group identifies the arithmetic homomorphism as well. These arguments prove the proper-target theorem for curves in Section 7. For a normal source, Section 8 uses a generic curve over a purely transcendental extension of K which preserves the connectedness of every finite geometric cover. The rational map so obtained extends to the source: a positive-dimensional fibre of its graph would contradict the existence of abelian covers of the target of arbitrarily large degree. Ramified covers at the cusps exclude the boundary of an open target. Finally, Section 9 applies the pro-p theorem on every finite pro-Σ cover, as in [12, Remark after Theorem 16.5], to recover the original pro-Σ homomorphism.

Notation and preliminaries

Until Section 9, put $\Pi_Z = \Pi_Z^{\{p\}}$ and $\Delta_Z = \Delta_Z^{\{p\}}$. For S an abelian variety or $\mathbf{G}_m$ over K, put $T_pS = \varprojlim S[p^n](\bar{K})$, with transition map $[p]$, and write $\kappa_{S,n} : S(L) \to H^1(L, S[p^n])$ for the Kummer map over a field extension L/K. In particular, $T_pS/p^nT_pS = S[p^n](\bar{K})$.

We use the correspondence between finite étalecovers and finite fundamental-group sets [5, Exposé V], the arithmetic exact sequence [5, Exposé IX, Théorème 6.1], and invariance of geometric fundamental groups under algebraically closed extension in characteristic zero [5, Exposé XIII, Corollaire 3.2 and Remarque 3.1.3]. For a smooth proper curve C with Jacobian J_C, the Abel–Jacobi identifications, formed using a geometric base point, are

$$\Delta_C^{\mathrm{ab}} = T_pJ_C, \qquad \mathrm{Lie}(J_C)^\vee = H^0(C, \omega_C). \tag{1.3}$$

Both identifications are independent of the base point and compatible with curve maps,

norms, and field extension; see [10, Propositions 2.2 and 9.1]. We use semistable reduction and the standard relative Picard construction, with references supplied in Section 4.

2 The ground field and descent of Tate homomorphisms

The valued overfield in Definition 1.1 serves only for the local construction. We first choose it with separably closed residue field, so that its tame Galois group is prime to p. We then show that a homomorphism of Tate modules is defined over a finite valued extension, even when K itself is not.

Lemma 2.1. (i) *In Definition 1.1, one may require a complete discretely valued field M with separably closed residue field. With a uniformizer π, one may then write*

$$M^{\mathrm{t}} = M(\pi^{1/n} : (n,p) = 1).$$

Every finite quotient of $\mathrm{Gal}(M^{\mathrm{t}}/M)$ has order prime to p.

(ii) *Subfields and finite extensions of p-subtame fields are p-subtame.*

(iii) *Finitely generated extensions of p-subtame fields are p-subtame.*

Proof. For a henselian discretely valued F, tame ramification gives $F^{\mathrm{t}} = F^{\mathrm{nr}}(\pi^{1/n} : (n,p) = 1)$ [16, Tags 04GK, 0EXW]. The completion of F^{nr} is a complete discretely valued field M with the same uniformizer and separably closed residue field. Compatible choices of roots embed F^{t} into M^{t}. All prime-to-p roots of unity lie in M by Hensel's lemma, so the finite root-adjoining stages are cyclic of prime-to-p order. Conversely, a complete discretely valued field is henselian.

Fix $K \subset M^{\mathrm{t}}$ and an algebraic closure Ω of M. A finite extension $K(\alpha)/K$ embeds into Ω, and $M(\alpha)/M$ is finite. Tame ramification is preserved by base change [16, Tag 0EXY]; hence $M(\alpha)M^{\mathrm{t}} \subset M(\alpha)^{\mathrm{t}}$. This proves (ii).

For independent variables $\mathbf{t}$, the Gauss completion of $M(\mathbf{t})$ is a complete discretely valued field of mixed characteristic $(0,p)$. By the same base-change assertion, $M^{\mathrm{t}}(\mathbf{t})$ embeds into its maximal tame extension. A finitely generated extension is finite over a purely transcendental one, proving (iii). □

Lemma 2.2 (Finite descent of Tate homomorphisms)**.** *Fix $K \subset M^{\mathrm{t}} \subset \Omega$ as above. Let S_1, S_2 be Jacobians or $\mathbf{G}_m$ over K, and choose a finite extension M_0/M inside M^{t} over which their defining equations descend. Every G_K-equivariant map $u : T_pS_1 \to T_pS_2$ is equivariant over the finite extension*

$$M_0(S_1[p], S_2[p]).$$

The same assertion holds for objects over finite extensions of K, using the same valued algebraic closure Ω.

Proof. The lattice $\mathrm{Hom}_{G_{M^{\mathrm{t}}}}(T_pS_1, T_pS_2)$ is saturated in $\mathrm{Hom}_{\mathbf{Z}_p}(T_pS_1, T_pS_2)$: if ph is invariant, then $p(\sigma h - h) = 0$ implies that h is invariant. Reduction modulo p therefore embeds its reduction into $\mathrm{Hom}_{\mathbf{F}_p}(S_1[p], S_2[p])$. Its Galois action factors through $\mathrm{Gal}(M^{\mathrm{t}}/M_0)$, a profinite group all of whose finite quotients have order prime to p.

Such an action on a $\mathbf{Z}_p$-lattice is detected modulo p. Indeed, if the lattice has rank r, then for every n the kernel of $\mathrm{GL}_r(\mathbf{Z}_p/p^n) \to \mathrm{GL}_r(\mathbf{F}_p)$ is a finite p-group, so its intersection with the finite prime-to-p image is trivial. The subgroup fixing both $S_1[p]$ and $S_2[p]$ consequently acts trivially on every $G_{M^{\mathrm{t}}}$-equivariant homomorphism. This applies to u, since $K \subset M^{\mathrm{t}}$.

For a finite extension $K(\alpha)/K$ inside Ω, replace M by $M(\alpha)$ and use $K(\alpha) \subset M(\alpha)^{\mathrm{t}}$. The residue field remains separably closed, and the same argument applies inside Ω. □

3 Finite covers and geometric conjugacy

The construction from differentials will first yield a morphism of curves over an algebraically closed field. To recover the prescribed homomorphism of fundamental groups, we shall apply that construction on finite covers, following the comparison preceding [12, Theorem 14.1]. The following profinite facts explain how equality on finite torsors gives equality up to a single geometric conjugation. They apply to every set of primes Σ.

Lemma 3.1. (i) *If D is a closed subgroup of a profinite group P, two continuous homomorphisms $\alpha, \beta : H \to P$ are D-conjugate if and only if their images in every finite quotient of P are conjugate by the image of D. It suffices to test a cofinal system of finite quotients.*

(ii) *An open subgroup of Π_Z^Σ is the relative pro-Σ fundamental group of the corresponding connected finite étalecover, viewed over its field of constants.*

(iii) *For a homomorphism $\theta : P_1 \to P_2$ over G, where $1 \to D_i \to P_i \to G \to 1$ is exact, its image is open if and only if $\theta(D_1)$ is open in D_2.*

Proof. For (i), the sets $\{d \in D : \beta \equiv d\alpha d^{-1} \pmod N\}$, for open normal $N \triangleleft P$, are nonempty closed subsets of the compact group D. They have the finite intersection property, and their intersection gives one conjugating element.

For (ii), let $\mathcal{D}$ be a profinite group and let $\mathcal{H}$ be the inverse image of an open subgroup of $\mathcal{D}^{(\Sigma)}$. Given an open normal $U \triangleleft \mathcal{H}$ with finite Σ-group quotient, take cores in $\mathcal{D}$. The quotient of the core of $\mathcal{H}$ by the core of U embeds into a finite product of conjugates of $\mathcal{H}/U$. Both this quotient and the quotient by the core of $\mathcal{H}$ are finite Σ-groups. Thus $\ker(\mathcal{D} \to \mathcal{D}^{(\Sigma)}) \subset U$. It follows that the maximal pro-Σ quotient of $\mathcal{H}$ is its image in $\mathcal{D}^{(\Sigma)}$. Apply this to the geometric fundamental group and then use the arithmetic exact sequence.

For (iii), the equalities $\theta(P_1) \cap D_2 = \theta(D_1)$ and $\theta(P_1)D_2 = P_2$ give $[P_2 : \theta(P_1)] = [D_2 : \theta(D_1)]$. □

In particular, passage to the cover corresponding to the image reduces an open homomorphism to a surjection.

A finite geometric cover need not be defined over the original ground field. The next observation allows a prescribed geometric Galois group to be realized after a finite extension of that field.

Lemma 3.2 (Realizing a geometric quotient)**.** *Let $1 \to D \to P \to \Gamma \to 1$ be exact, and let $N \triangleleft D$ be open. After restricting to an open normal subgroup of Γ, the quotient map $D \to D/N$ extends to a quotient of the restricted extension.*

Proof. Choose an open normal $B \triangleleft P$ with $B \cap D \subset N$. The subgroup DB is the inverse image of an open normal subgroup of Γ. Since $[B, D] \subset B \cap D \subset N$, the subgroup NB is normal in DB, and the inclusion of D induces $D/N \simeq DB/NB$. □

Lemma 3.3 (Arithmetic torsor comparison)**.** *Let $\theta : \Pi_X^\Sigma \to \Pi_Y^\Sigma$ be over G_K, and let $f : X \to Y$ be a K-morphism. For an open normal $N \triangleleft \Pi_Y^\Sigma$, let $V_N \to Y$ be the corresponding Π_Y^Σ/N-torsor, and let $W_N \to X$ be the pullback by θ. If, for a cofinal system of N, there is a Π_Y^Σ/N-equivariant isomorphism*

$$W_N \simeq X \times_Y V_N$$

inducing the identity on the common constant-field quotient, then f_* and θ differ by one inner automorphism from Δ_Y^Σ.

Proof. The isomorphism of finite fibres conjugates the two monodromy actions by an element of Π_Y^Σ/N. Its identity action on the constant quotient says that this element lies in the image of Δ_Y^Σ. Apply Lemma 3.1(i). □

4 Integral realization and finite Kummer classes

The passage from Tate modules to formal groups and geometric cohomology classes is already present in Mochizuki's proof [12, §6 and the proof of Proposition 7.4]; see also Faltings's account in [4, Proposition 11]. We need an integral version at each finite Kummer level over a discrete valuation ring with arbitrary residue field. We obtain it by characterizing the connected Tate lattice by inertia and applying Tate's full faithfulness theorem. The differential of the resulting formal homomorphism will then supply the map on regular differentials.

Throughout, R_0 is a complete discrete valuation ring of mixed characteristic $(0,p)$, K_0 is its fraction field, and Ω is an algebraic closure of K_0, endowed with the extended valuation. The residue field of R_0 is arbitrary.

4.1 Semistable models and their connected torsion

Let C/K_0 be a smooth proper geometrically connected curve. By semistable reduction [1, §9.2, Theorem 7], after a finite extension of K_0 we may choose a regular split semistable model $\mathcal{C}/R_0$. The stable reduction theorem in genus at least two is due to Deligne and Mumford [2, Corollary 2.7]. A further finite extension gives a section through the smooth locus [1, §2.3, Proposition 5]. Put

$$J = \mathrm{Pic}^0_{C/K_0}, \qquad \mathcal{G} = \mathrm{Pic}^0_{\mathcal{C}/R_0}\,.$$

Here Pic^0 denotes the subgroup of classes of multidegree zero. Raynaud's theorem identifies $\mathcal{G}$ with the identity component of the Néron model of J [15, Théorèmes 8.1.4(b) and 8.2.1]; we use the formulation in [1, §9.5, Theorem 4(b) and Remark 5]. It is smooth, separated, and of finite type, with generic fibre J. Its special fibre is an extension of the product of the Jacobians of the normalized components by a split torus [1, §9.2, Example 8]. Thus $\mathcal{G}$ is semiabelian, and its formation commutes with faithfully flat extensions of discrete valuation rings [1, §7.4, Corollary 4].

Write $\widehat{\mathcal{G}}$ for the formal completion at the identity. Multiplication by p^n on $\mathcal{G}$ is faithfully flat and quasi-finite. Its local factor at the identity of the special fibre is a finite flat open and closed subgroup

$$\widehat{\mathcal{G}}[p^n] \subset \mathcal{G}[p^n].$$

These kernels form the connected p-divisible group associated to $\widehat{\mathcal{G}}$ [18, §2.2, Proposition 1]. We shall also use the same notation for $S = \mathbf{G}_m$, with $\mathcal{G} = \mathbf{G}_{m,R_0}$, whose formal p^n-torsion is μ_{p^n}.

The following description of $T_p\widehat{\mathcal{G}}$ makes it functorial for Galois equivariant homomorphisms of Tate modules.

Lemma 4.1 (The connected Tate module)**.** *Let S be one of these Jacobians or $\mathbf{G}_m$, with model $\mathcal{G}$. Then $T_p\widehat{\mathcal{G}}$ is saturated in T_pS, the quotient $T_pS/T_p\widehat{\mathcal{G}}$ is unramified, and*

$$T_p\widehat{\mathcal{G}} = T_pS \cap \sum_{\sigma \in I_{K_0}} (\sigma - 1)(T_pS \otimes_{\mathbf{Z}_p} \mathbf{Q}_p).$$

In particular, every G_{K_0}-equivariant map between such Tate modules preserves their connected lattices. A rationally surjective map is rationally surjective on the connected lattices as well.

Proof. The injections $\widehat{\mathcal{G}}[p^n](\Omega) \hookrightarrow S[p^n](\Omega)$ show that $T_p\widehat{\mathcal{G}}$ is saturated in T_pS. To prove that the quotient is unramified, fix $\tau \in J[p^n](\Omega)$ and $\sigma \in I_{K_0}$. Represent τ by a degree-zero divisor, and choose a finite Galois extension over which all its points are rational. A point z and its conjugate σz have the same specialization. If this specialization is smooth, the

difference of their horizontal divisors restricts to the trivial line bundle on the special fibre. In nodal coordinates, the function

$$\frac{x - x(\sigma z)}{x - x(z)}$$

defines the same horizontal difference near the node, since the vertical orders cancel. On the normalized special branches its reductions are nonzero constants. Summing these differences gives an extension of $\sigma\tau - \tau$ whose restriction to each normalized component is trivial. Its p^n-th power is trivial by separatedness of $\mathcal{G}$. Its reduction is therefore a p^n-torsion point of the gluing torus, hence the identity in characteristic p. It follows that $\sigma\tau - \tau \in \widehat{\mathcal{G}}[p^n](\Omega)$, as required. For $\mathbf{G}_m$ the formal and full Tate modules coincide.

It remains to show that the inertia differences span $T_p\widehat{\mathcal{G}} \otimes \mathbf{Q}_p$. More generally, the rational Tate module of a connected p-divisible group over R_0 has no nonzero unramified quotient. The image of the integral Tate module in such a quotient is a Galois stable lattice, hence the Tate module of an étale$p$-divisible group over R_0 [18, §1.4]. Tate's full faithfulness theorem extends the quotient map to a homomorphism over R_0 [18, §4.2, Theorem 4]. This homomorphism is zero. Indeed, after strict henselization each finite level of the source is connected, whereas the target is a disjoint union of copies of the base. The quotient of $T_p\widehat{\mathcal{G}} \otimes \mathbf{Q}_p$ by the span of its inertia differences is unramified, and must therefore vanish. Since $T_pS/T_p\widehat{\mathcal{G}}$ is unramified, every inertia difference in $T_pS \otimes \mathbf{Q}_p$ lies in $T_p\widehat{\mathcal{G}} \otimes \mathbf{Q}_p$. Saturation gives the displayed formula. An equivariant homomorphism preserves inertia differences, and a surjective homomorphism maps their span onto the corresponding span in the target. This proves the final assertions. □

Proposition 4.2 (Tate realization)**.** *Let S_i and $\mathcal{G}_i$ be as above. A G_{K_0}-equivariant homomorphism $u : T_pS_1 \to T_pS_2$ induces a unique formal group homomorphism over R_0*

$$F_u : \widehat{\mathcal{G}}_1 \longrightarrow \widehat{\mathcal{G}}_2, \qquad T_pF_u = u|_{T_p\widehat{\mathcal{G}}_1}.$$

The assignment $u \mapsto dF_u$ is additive and compatible with composition and finite extensions. After base change to Ω, the induced map on Lie algebras is independent of the semistable models and finite local stages used in its construction. For an algebraic homomorphism f defined after a finite extension of K_0, one has $dF_{T_pf} = df$ over that extension. Moreover, $dF_u \neq 0$ if either $u \otimes \mathbf{Q}_p$ is surjective and $\dim S_2 > 0$, or $S_1 = \mathbf{G}_m$ and $u \neq 0$.

Proof. By Lemma 4.1, u restricts to the connected lattices. Tate's theorem extends this restriction uniquely to a homomorphism of p-divisible groups, and hence of their associated formal groups [18, §4.2, Theorem 4 and Corollary 1; §2.2, Proposition 1]. Uniqueness gives additivity and compatibility with composition. The identification with the semistable Néron identity component, and its compatibility with base change, show that the construction is independent of the chosen model and finite stage. An algebraic homomorphism extends by the Néron mapping property; its formal completion induces the prescribed Tate map and therefore coincides with F_{T_pf}. Taking derivatives proves $dF_{T_pf} = df$.

If $u \otimes \mathbf{Q}_p$ is surjective, Lemma 4.1 shows that its restriction to the connected part is surjective over $\mathbf{Q}_p$. This target is nonzero when $\dim S_2 > 0$. If $S_1 = \mathbf{G}_m$, the connected Tate module is the whole Tate module. Thus either hypothesis implies $F_u \neq 0$. The formal logarithm in characteristic zero then gives $dF_u \neq 0$. □

4.2 Correcting points by algebraic constants

Mochizuki translates points by constants to place them in the identity component of a Néron model in the argument preceding [12, Lemma 9.2]. For the valued extensions used here, we

make this correction directly on divisors of the semistable curve. The correcting point lies in $S(\Omega)$ and is therefore divisible by every power of p; its subtraction does not change a finite Kummer class.

Lemma 4.3 (Points modulo constants)**.** *Let F/K_0 be an extension of complete discretely valued fields whose valuation ring dominates R_0, and let E/F be algebraic with $\Omega \subset E$. For a model $\mathcal{G}$ as above, every $P \in S(E)$ admits an expression*

$$P = z + c, \qquad z \in \mathcal{G}(\mathcal{O}_E), \quad c \in S(\Omega).$$

For $\mathbf{G}_m$ the displayed sum is interpreted multiplicatively.

Proof. For $S = \mathbf{G}_m$, choose $c \in \Omega^\times$ with $v(c) = v(P)$; this is possible since the values are rational multiples of $v(p)$. Then $P/c \in \mathcal{O}_E^\times$.

Now suppose $S = J$. Choose a finite extension F'/F inside E over which P is defined. The section of $\mathcal{C}$ allows us to represent P by a degree-zero line bundle, and hence by a divisor $\sum m_Z Z$ on $C_{F'}$ [1, §8.1, Proposition 4]. It suffices to choose constant points $c_Z \in C(\Omega)$ for which

$$\overline{Z} - [F'(Z) : F']\,\overline{c_Z}$$

is Cartier of multidegree zero. We may enlarge F' by finite constant extensions inside E to make the chosen c_Z rational. If such an extension splits Z, the same choice of constant point is retained for each point above it.

If Z specializes to the smooth locus, take c_Z specializing to a smooth point of the same component; Hensel's lemma gives such a point over a finite constant extension. The closure of Z is Cartier and has special degree $[F'(Z) : F']$ on that component and degree zero on the others. The required difference therefore has multidegree zero.

Suppose instead that Z specializes to a node. In a split étalechart the completed local ring is $\mathcal{O}_{F'}[[x, y]]/(xy - q)$, with $q \in R_0$. Normalize v to have value group $\mathbf{Z}$ on F', and put $a = x(Z)$. Then $0 < v(a) < v(q)$ and $F'(Z) = F'(a)$: the other coordinate is q/a, and the lift through the étalechart is uniquely determined by Hensel's lemma. Write $d = [F'(Z) : F']$, and let P_a be the minimal polynomial of a. The horizontal divisor of $P_a(x)$ is $\overline{Z}$, with multiplicity one. Since all conjugates of a have valuation $v(a)$, the vertical order along the branch $x = 0$ is $v(N_{F'(a)/F'}(a)) = d\,v(a)$; along the other branch it is zero. Choose $a_c \in \Omega$ with $v(a_c) = v(a)$ and lift $(a_c, q/a_c)$ through the chart to obtain c_Z. The quotient

$$\frac{P_a(x)}{(x - a_c)^d}$$

is a local equation for $\overline{Z} - d\,\overline{c_Z}$. On the normalized special branches it reduces to 1 and $\overline{N_{F'(a)/F'}(a)/a_c^d}$, respectively, after canceling the vertical orders. Both are nonzero constants. The associated line bundle is therefore trivial on each normalized component, and in particular has multidegree zero. Being Cartier can be checked after the faithfully flat completion; away from the node it is already clear. The same calculation applies after a constant extension, since the component and the valuation specifying c_Z are unchanged.

The sum of these corrected divisors is Cartier of multidegree zero, and its generic class is $P - [\sum m_Z [F'(Z) : F'] c_Z]$. It defines a point of $\mathcal{G}(\mathcal{O}_{F'})$, and hence the required point of $\mathcal{G}(\mathcal{O}_E)$. □

Lemma 4.4 (Preservation of Kummer classes)**.** *Retain the preceding notation and assume that the residue field of E is separably closed. For $u : T_pS_1 \to T_pS_2$ as in Proposition 4.2, $P \in S_1(E)$, and $n \geq 1$, there is $Q_n \in S_2(E)$ such that*

$$u_n \kappa_{S_1,n}(P) = \kappa_{S_2,n}(Q_n) \quad \text{in } H^1(E, S_2[p^n]).$$

Proof. Write $P = z + c$ as in Lemma 4.3. Since $S_1(\Omega)$ is p-divisible, it is enough to treat the integral point z. Work over the strictly henselian ring $\mathcal{O}_E$. The quotient of $\mathcal{G}_i[p^n]$ by its open subgroup $\widehat{\mathcal{G}}_i[p^n]$ is étale. Smooth-group torsors over a strictly henselian local ring are trivial [3, Exposé XXIV, Proposition 8.1 and its footnote]. The fppf cohomology sequences show that every $\mathcal{G}_i[p^n]$-torsor lifts to a $\widehat{\mathcal{G}}_i[p^n]$-torsor and is itself the division torsor of a point of $\mathcal{G}_i(\mathcal{O}_E)$.

Lift the division torsor of z to the connected subgroup $\widehat{\mathcal{G}}_1[p^n]$. Push it out by $F_u[p^n]$ and then by the inclusion into $\mathcal{G}_2[p^n]$. The resulting torsor is the division torsor of some $Q_n \in \mathcal{G}_2(\mathcal{O}_E)$. On generic fibres, $F_u[p^n]$ is the restriction of u_n, so this construction gives the asserted equality of Kummer classes. □

4.3 The differential estimate

The integral comparison between Kummer classes and differentials is the step carried out by p-adic Hodge theory in [12, §9, before Lemma 9.2]. Here it follows from the formal homomorphism and the point correction above: equality of finite Kummer classes gives divisibility of the difference of the logarithmic derivatives in the integral Lie algebra of the fixed model.

For a derivation D of an R_0-algebra and a point z of a smooth commutative group, write $\partial_D z$ for its logarithmic derivative, characterized by $\langle \eta, \partial_D z \rangle = \langle z^*\eta, D \rangle$ for invariant differentials η. This operation is additive and is natural under group homomorphisms.

Lemma 4.5 (Integral differential comparison)**.** *Let u and F_u be as in Proposition 4.2, and let F/K_0 and E/F be as in Lemma 4.3. Suppose $D : E \to E$ is a K_0-derivation preserving $\mathcal{O}_E$, with $D(F) \subset F$ and continuous restriction to F. For $x \in \widehat{\mathcal{G}}_1(\mathfrak{m}_F)$ and every $n \geq 1$,*

$$\kappa_{S_2,n}(F_u(x)) = u_n \kappa_{S_1,n}(x), \qquad \partial_D(F_u(x)) = dF_u(\partial_D x).$$

The first equality is in $H^1(F, S_2[p^n])$. If $S_2 = J$ and $z \in J(E)$ has $\kappa_{J,n}(z) = u_n \kappa_{S_1,n}(x)$, then

$$\partial_D z - dF_u(\partial_D x) \in p^n \operatorname{Lie}(\mathcal{G}_2) \otimes_{R_0} \mathcal{O}_E.$$

Proof. The series F_u converges at x, with $F_u(x) \in \widehat{\mathcal{G}}_2(\mathfrak{m}_F)$. Finite flatness of $[p^n]$ gives a division point y of x in the formal neighborhood over a finite extension of F. For $\sigma \in G_F$ one has

$$\sigma F_u(y) - F_u(y) = F_u(\sigma y - y) = u_n(\sigma y - y).$$

This proves the equality of Kummer classes. Naturality of invariant differentials, together with continuity of D, gives the equality of logarithmic derivatives.

For the final assertion, the Kummer exact sequence gives $z - F_u(x) = [p^n]w$ with $w \in J(E)$. Apply Lemma 4.3 to write $w = w_0 + a$, where $w_0 \in \mathcal{G}_2(\mathcal{O}_E)$ and $a \in J(\Omega)$. Since a K_0-derivation vanishes on Ω,

$$\partial_D z - dF_u(\partial_D x) = p^n \partial_D w_0.$$

Finally, $\partial_D w_0$ belongs to $\operatorname{Lie}(\mathcal{G}_2) \otimes_{R_0} \mathcal{O}_E$ because D preserves $\mathcal{O}_E$. This proves the inclusion. □

5 From finite Kummer classes to curve points

To pass from points of Jacobians to points of curves, we follow the Picard-torsor argument of [12, §4, Proposition 7.4 and §8] and its presentation in [4, §5, Proposition 9 and Definition 10; §6]. A sufficiently divisible lift of a section trivializes the degree-one Picard torsor. After passing to a cover of suitable genus, the pro-p hypothesis on the absolute Galois group turns

a degree-one Picard class into a rational point. We obtain points whose sections agree with a prescribed section in a fixed finite Kummer quotient.

For a proper curve C with a chosen geometric base point, $\alpha_C : C \to J_C$ denotes the corresponding Abel–Jacobi morphism, defined over the field of that point. For $\mathbf{G}_m$, the notation $\alpha_{\mathbf{G}_m}$ denotes the identity. The differential pullback in the proper case is independent of the base point, since translations on a Jacobian preserve invariant differentials.

Lemma 5.1 (A divided torsor)**.** *Let A/F be an abelian variety in characteristic zero, let T be an A-torsor, and let $s : G_F \to \Pi_T$ be a section. Let m be a power of p. Suppose E/F is finite of degree dividing m and $P \in T(E)$ has the same $A[m]$-conjugacy class of sections as $s|_{G_E}$ in the quotient whose geometric kernel is $A[m]$. Then $T(F) \neq \varnothing$.*

Proof. The cocycle description of the extension with geometric kernel $A[m]$ associates to the splitting induced by s a class $\eta \in H^1(F, A)$ with $m\eta = [T]$ [7, Proposition 2.2 and Lemma 2.3]. In this description, agreement with the point section of P means that $\eta|_E = 0$. Restriction followed by corestriction gives $[E : F]\eta = 0$. Since $[E : F] \mid m$, it follows that $[T] = m\eta = 0$. □

Lemma 5.2 (Degree-one Picard classes over pro-p fields)**.** *Let F have characteristic zero and pro-p absolute Galois group. Let C/F be a smooth proper geometrically connected curve of genus g. If $\mathrm{Pic}^1_C(F) \neq \varnothing$ and either p is odd or g is odd, then $C(F)$ is nonempty. In fact its points avoid any prescribed finite closed subscheme.*

Proof. Let $\xi \in \mathrm{Pic}^1_C(F)$, and let $\beta \in \mathrm{Br}(F)$ be the obstruction to representing ξ by a line bundle over F [1, §8.1, Proposition 4]. Choose a degree-one representative M and descent isomorphisms over a finite Galois extension F'/F, whose scalar cocycle $c_{\sigma,\tau}$ represents β. This extension has p-power degree, so restriction–corestriction shows that β has p-power order.

We use the determinant of cohomology to obtain an annihilator prime to p; this is the Picard-obstruction argument recalled by Faltings [4, p. 143]. Choose $m > 2g - 2$ with $p \nmid m(m + 1 - g)$, which is possible under the stated parity hypothesis. Riemann–Roch gives $h^0(C_{F'}, M^{\otimes m}) = m + 1 - g$. Taking determinants of the semilinear maps on this space of sections makes the scalar cocycle $c_{\sigma,\tau}^{m(m+1-g)}$ a coboundary. Hence $m(m + 1 - g)\beta = 0$, and $\beta = 0$. Thus ξ is represented by a degree-one line bundle L over F.

For a sufficiently large d prime to p, the line bundle $L^{\otimes d}$ has a section whose zero divisor avoids the prescribed finite subscheme: the evaluation map is surjective by Riemann–Roch, and F is infinite. The resulting effective divisor has degree d. Every closed point has p-power degree, so at least one point in its support has degree one. □

We now apply these observations to a twist of an abelian cover of the target. The use of covers carrying a lift of a section is the method underlying Tamagawa's criterion for geometric sections [17, Corollary 2.10]. Here we require agreement with the section only modulo a prescribed power of p; the pro-p hypothesis and the preceding two lemmas provide the required rational point.

Lemma 5.3 (Finite point approximation)**.** *Let X/K be a smooth proper geometrically connected curve or $\mathbf{G}_m$, let Y/K be proper hyperbolic, and let $\theta : \Pi_X \to \Pi_Y$ be over G_K. Let $L \supset \bar{K}$ have pro-p absolute Galois group. Suppose that on every finite geometric p-cover induced by θ, the corresponding Tate map preserves finite Kummer images over every finite extension of L. Then, for every $x \in X(L)$ and $n \geq 1$, there is $y_n \in Y(L)$ whose point section agrees with θs_x after geometric abelianization modulo p^n.*

Proof. Put $s = \theta s_x$ and choose $a \geq \max(n, 2)$. A point of $Y(\bar{K})$ supplies a splitting, and G_L acts trivially on T_pJ_Y/p^a. Thus the abelianized extension modulo p^a is $(T_pJ_Y/p^a) \times G_L$. Write $s(g) = (\rho_a(g), g)$. The quotient $(q, g) \mapsto q - \rho_a(g)$ defines a geometrically connected (T_pJ_Y/p^a)-torsor $Y_a^s \to Y_L$ to which s lifts. The Abel–Jacobi morphism from Y_a^s to its degree-one Picard torsor therefore carries the lifted section to a section of $\mathrm{Pic}^1_{Y_a^s/L}$.

Let T/L be the finite extension cut out by $\ker \rho_a$. Over T the twisted cover becomes a geometric cover V, and x lifts to a point z of the corresponding pullback cover U. Choose all covers and their constant base points over a finite extension of K inside $\bar{K}$. Then U is a proper curve or, in the multiplicative case, a copy of $\mathbf{G}_m$ with the chosen base point corresponding to 1. These choices identify the Picard torsor with J_V over T. The arithmetic base-point cocycle vanishes on G_T, since T contains $\bar{K}$. Relative to the chosen constant point of Pic^1_V, the difference of sections at level p^m is consequently the image of $\kappa_{J_U,m}(\alpha_U(z))$ in the proper case, or $\kappa_{\mathbf{G}_m,m}(z)$ in the multiplicative case, under the homomorphism induced by θ on geometric abelianizations. This is the section-difference calculation of [17, Lemma 2.6], applied to the finite p-primary quotient.

Take m with $[T : L] \mid p^m$. By the Kummer-image hypothesis this difference is the Kummer class of a point of $J_V(T)$. Translating the constant Picard point by that point gives a point whose section agrees with the prescribed section modulo p^m. Applying Lemma 5.1 to the Picard torsor over L gives $\mathrm{Pic}^1_{Y_a^s/L}(L) \neq \varnothing$. Riemann–Hurwitz gives

$$g(Y_a^s) - 1 = p^{2g(Y)a}(g(Y) - 1).$$

Thus its genus is odd when $p = 2$, and Lemma 5.2 produces an L-point of Y_a^s. Its image y_n has the required section difference modulo p^a, and hence modulo p^n. □

6 Reconstruction from differentials

The reconstruction of a curve morphism by preserving relations among regular differentials is the argument of Mochizuki [12, Proposition 12.1 and Corollary 13.4]; see also Faltings [4, §6]. The points just constructed need not be compatible as n varies. Their differential pullbacks nevertheless approach a fixed linear map, and this will suffice to reconstruct a morphism. The reason is that a proper hyperbolic curve is recovered from ratios of regular differentials together with their first derivatives. The derivatives retain the information lost by the canonical map in the hyperelliptic case. A derivation bounded for the valuation then allows the algebraic relations among these functions to pass to the limit.

6.1 Canonical ratios and their first derivatives

Lemma 6.1. *Let C/k be a proper hyperbolic curve in characteristic zero, with basis $\omega_0, \dots, \omega_{g-1}$ of its regular differentials. Set*

$$U = \{\omega_0 \neq 0\}, \qquad r_i = \omega_i/\omega_0, \qquad h_i = dr_i/\omega_0.$$

Then U is affine and

$$\mathcal{O}(U) = k[r_i]_i + \sum_i k[r_j]_j h_i = k[r_i, h_i]_i. \tag{6.1}$$

Consequently dominant rational maps from an integral k-scheme to C are determined by their pullbacks of regular differentials.

Proof. The canonical map is an embedding unless C is geometrically hyperelliptic [12, Lemma 10.4(1)–(2)]. In the hyperelliptic case, uniqueness of the involution makes it descend

to k. Its quotient is a smooth genus-zero curve, and the canonical map factors through this quotient, followed by the embedding defined by a complete linear system of degree $g-1$. The inverse image of the affine canonical chart is U, so $\mathcal{O}(U)$ is finite over $k[r_i]_i$. In the nonhyperelliptic case, $\mathcal{O}(U) = k[r_i]_i$.

In the hyperelliptic case, the quotient has genus zero, so the involution ι acts as -1 on regular differentials. Its invariant ring is $k[r_i]_i$, and $\mathcal{O}(U) = k[r_i]_i \oplus \mathcal{O}(U)^-$, where the second summand is the anti-invariant part. Invariant differentials on U are precisely the pullbacks of differentials from $\operatorname{Spec} k[r_i]_i$. Indeed, away from the branch locus this follows from étaledescent; at a branch point, the local equation $z^2 = b$ gives $z\,dz = db/2$ and proves the same assertion. Since ω_0 trivializes the differentials on U, their invariant part is $\mathcal{O}(U)^-\omega_0$. The coordinate differentials $dr_i = h_i\omega_0$ generate this module over $k[r_i]_i$. Thus $\mathcal{O}(U)^- = \sum_i k[r_j]_j h_i$, which proves (6.1).

For the last assertion, two dominant maps with the same differential pullback have the same values on every r_i. Differentiating and dividing by the common nonzero pullback of ω_0 gives the same values on every h_i. Equation (6.1) proves equality of the maps on the generic point. □

Proposition 6.2 (Reconstruction of a morphism)**.** *Let $k \subset E \subset L$ be characteristic-zero fields with E/k finitely generated of transcendence degree one. Let v be a real-valued nonarchimedean valuation on L, and let $D : L \to L$ be a derivation satisfying*

$$D(k) = 0, \quad D(E) \subset E, \quad D|_E \neq 0, \qquad v(Db) \geq v(b).$$

Let C/k be proper hyperbolic and let $a : H^0(C, \omega_C) \to \Omega^1_{E/k}$ be a nonzero k-linear map. Suppose that for every N there exists $y_N \in C(L)$ such that, on one basis ω_i,

$$v\big(\langle y_N^*\omega_i, D\rangle - \langle a(\omega_i), D\rangle\big) \geq N.$$

Then there is a unique nonconstant $q \in C(E)$ with $q^* = a$.

Proof. Evaluation at D identifies $\Omega^1_{E/k}$ with E. Put $A_i = \langle a(\omega_i), D\rangle$ and reorder the basis so that $A_0 \neq 0$. Write $\beta_{i,N} = \langle y_N^*\omega_i, D\rangle$. For large N, $\beta_{0,N} \neq 0$, and hence $y_N \in U(L)$, with U as in Lemma 6.1. Its evaluation map $\varphi_N : \mathcal{O}(U) \to L$ satisfies

$$\varphi_N(r_i) = \frac{\beta_{i,N}}{\beta_{0,N}}, \qquad \varphi_N(h_i) = \frac{D\varphi_N(r_i)}{\beta_{0,N}}.$$

The denominators have eventually constant valuation, and D does not decrease the valuation of an error. Therefore these values converge, in the valuation sense, to

$$A_i/A_0, \qquad D(A_i/A_0)/A_0 \quad \text{in } E.$$

Every polynomial relation among the r_i, h_i remains true at these limits. Indeed, evaluation of a polynomial is continuous for the valuation, and all the proposed limits already belong to E; no completeness assumption on E or L is involved. Equation (6.1) therefore gives a k-algebra map $\psi : \mathcal{O}(U) \to E$ and a point $q \in U(E)$. For every $b \in \mathcal{O}(U)$, both $\varphi_N(b) \to \psi(b)$ and $D\varphi_N(b) \to D\psi(b)$.

Write each regular differential on U as a finite sum $\omega_i = \sum_j c_{ij}\,db_j$, where b_j run through the generators r_i, h_i and $c_{ij} \in \mathcal{O}(U)$. Passing to the limit in this expression shows $\langle q^*\omega_i, D\rangle = A_i$. Thus $q^* = a$; in particular q is nonconstant. Uniqueness follows from Lemma 6.1. □

6.2 A valued generic point

To apply Proposition 6.2, we realize the generic point of the source curve in a valued field on which differentiation is bounded. We also need the residue field to be separably closed and the absolute Galois group to be pro-p, as required by the approximation argument. The following construction provides these properties simultaneously. The compositum of an unramified extension with the algebraic constants is the valued field used by Faltings [4, §6, pp. 146–147]. We record the boundedness of a derivation on this field, which will allow us to pass algebraic relations to the limit.

Lemma 6.3. *Let K_0 be a complete discretely valued field of mixed characteristic $(0,p)$ and let X/K_0 be a smooth geometrically connected curve. After a finite extension of K_0, its function field embeds in $F = \widehat{K_0(t)}$, completed for the Gauss valuation, so that its generic point lies in any prescribed formal neighborhood of a chosen rational smooth point. On*

$$L = F^{\mathrm{nr}}\Omega, \qquad \Omega = \overline{K_0},$$

the derivation $D = \partial/\partial t$ satisfies

$$D(\Omega) = 0, \quad D(K_0(X)) \subset K_0(X), \quad D|_{K_0(X)} \neq 0, \quad v(Db) \geq v(b). \tag{6.2}$$

The field L and all its finite extensions have separably closed residue fields, value group $\mathbf{Q}$ when $v(p) = 1$, and pro-p absolute Galois groups.

Proof. Choose a rational smooth point after finite extension, an étalecoordinate z vanishing there, and a standard étaleequation $P(z,y) = 0$ normalized by $P_y(0,0) = 1$. With a uniformizer π and sufficiently large m,

$$\pi^{-m} P(\pi^{2m}t, \pi^m w) = w + \pi H(t,w), \qquad H \in \mathcal{O}_{K_0}[t,w].$$

Hensel's lemma gives a point over F with $z = \pi^{2m}t$. It is generic, and m can be increased to place it in any prescribed formal neighborhood. Differentiation does not decrease Gauss valuation on $K_0(t)$, and consequently extends continuously to F. Separability implies that it preserves the embedded function field.

For each finite constant extension K_1/K_0, the field FK_1 is the Gauss completion of $K_1(t)$, so the same estimate holds. Formal étaleness gives unique extension of a derivation across a finite étalealgebra [16, Tag 00UR]. Applying this to the unramified extensions, after constant extension, shows that D preserves $\mathcal{O}_L$. Every value in L is realized by a constant in Ω. Dividing an element by a constant of the same value proves the full inequality in (6.2).

The residue field of L is separably closed. Since L/F is algebraic and Ω contains all roots of p, its value group is $\mathbf{Q}$; these statements persist under finite extension. For every prime $\ell \neq p$, each such extension T has ℓ-divisible multiplicative group: divide the value by ℓ, then apply Hensel's lemma to a unit. Kummer theory excludes cyclic extensions of degree ℓ over every finite extension of L. A non-p finite Galois quotient would contain a subgroup of order ℓ and yield just such an extension. Hence G_L is pro-p. □

6.3 The geometric curve morphism

Let S_1, S_2 be Jacobians or $\mathbf{G}_m$ over a p-subtame field K, and let $u : T_pS_1 \to T_pS_2$ be G_K-equivariant. Fix an embedding $K \subset M^{\mathrm{t}} \subset \Omega$ as in Section 2. Lemma 2.2 and Proposition 4.2 define an Ω-linear differential

$$\lambda_u : \mathrm{Lie}(S_1)_\Omega \longrightarrow \mathrm{Lie}(S_2)_\Omega. \tag{6.3}$$

It is compatible with extension of the finite field of definition, composition, and algebraic group morphisms. The proposition below constructs a geometric curve morphism from this differential. Its descent to K, and the comparison with the original nonabelian homomorphism, will be proved in the next section.

Proposition 6.4. *Let K be p-subtame, X/K a smooth proper geometrically connected curve or $\mathbf{G}_m$, and Y/K proper hyperbolic. Choose geometric base points on the proper curves, and use 1 on $\mathbf{G}_m$. Let $\theta : \Pi_X \to \Pi_Y$ be over G_K. Its geometric abelianization θ^{ab} has target T_pJ_Y and source T_pJ_X in the proper case, or $\mathbf{Z}_p(1)$ when $X = \mathbf{G}_m$. If*

$$a_\theta = \alpha_X^* \lambda_{\theta^{\mathrm{ab}}}^* : H^0(Y_\Omega, \omega_{Y_\Omega}) \longrightarrow H^0(X_\Omega, \Omega^1_{X_\Omega/\Omega})$$

is nonzero, there is a unique nonconstant morphism $f_\Omega : X_\Omega \to Y_\Omega$ with $f_\Omega^* = a_\theta$. The construction commutes with morphisms of the relevant finite covers and their geometric deck transformations.

Proof. Descend the curves, base points, and θ^{ab} to a finite extension M_0/M of the local field using Lemma 2.2. Enlarge M_0 to choose the semistable models and the integral formal map $F_{\theta^{\mathrm{ab}}}$. Apply Lemma 6.3 near a base point of X so that its generic point $x \in X(F)$ has Abel–Jacobi image $\alpha_X(x)$ in the formal neighborhood of the identity. Set $L = F^{\mathrm{nr}}\Omega$ and $q = F_{\theta^{\mathrm{ab}}}(\alpha_X(x)) \in J_Y(F)$. Lemma 4.5 identifies the Kummer class of q modulo p^n with the image under $(\theta^{\mathrm{ab}})_n$ of the Kummer class of $\alpha_X(x)$, taken in J_X or $\mathbf{G}_m$ according to the source. It also gives

$$\partial_D q = \lambda_{\theta^{\mathrm{ab}}}(\partial_D \alpha_X(x)).$$

To obtain points of Y approximating q, we check the hypothesis of Lemma 5.3 over this same field L. The untwisted covers V, U occurring there and their induced Tate maps descend to a finite extension of K. Lemma 2.2 puts these data and their models over some finite M_1/M_0 inside Ω. For any finite T/L, the field $FM_1 \subset T$ is a complete discretely valued field and T/FM_1 is algebraic with separably closed residue field. Thus Lemma 4.4 preserves every finite Kummer image over T itself. Finite geometric p-covers of $\mathbf{G}_m$ become $t \mapsto t^{p^b}$ after finite constant extension [10, Example 9.8], so the same argument includes that case.

Lemma 5.3 now supplies $y_n \in Y(L)$ whose Abel–Jacobi images have the same Kummer classes as q modulo p^n. Apply Lemma 4.5, using the one fixed lattice of invariant differentials on the model of J_Y. Their pullbacks to Y form a basis by (1.3). We obtain the approximation required in Proposition 6.2, with $M_0 \subset M_0(X) \subset L$. This gives a nonconstant rational map over M_0, which extends because the source is a smooth curve and the target is proper [16, Tag 0BXZ]. Base change gives f_Ω.

Uniqueness follows from Lemma 6.1. For a comparison between covers, all the data can be defined over a common finite extension of the local field. Naturality of $\lambda_{\theta^{\mathrm{ab}}}$, followed by the same uniqueness, proves the assertions about finite covers and geometric deck maps. □

7 The Hom theorem for proper curves

The preceding section constructs a geometric morphism with the differential prescribed by θ. We must still prove that it induces θ on the fundamental group. We follow Mochizuki's comparison over finite covers [12, §14, proof of Theorem 14.1], also used by Faltings [4, proof of Theorem 14]. The construction is applied to every finite cover of the target. Its uniqueness makes the resulting morphisms compatible and equivariant under deck transformations, so that they recover the full geometric homomorphism. The following lemma then gives descent to the ground field and equality on the arithmetic fundamental groups. We include the descent and centralizer arguments for the present ground fields.

Lemma 7.1 (Geometric faithfulness and arithmetic descent)**.** *Let X/k be a smooth proper geometrically connected curve and Y/k proper hyperbolic, in characteristic zero. A nonconstant morphism $X \to Y$ is determined by its geometric outer pro-p homomorphism. If a nonconstant morphism over an algebraically closed extension Ω/k has the same geometric outer homomorphism as an open $\theta : \Pi_X \to \Pi_Y$ over G_k, it descends to a unique k-morphism, whose entire arithmetic homomorphism is θ modulo geometric conjugation.*

Proof. The Tate-module functor is faithful on abelian varieties [9, Lemma 12.2]. Thus equal geometric outer homomorphisms give equal Jacobian homomorphisms and equal differential pullbacks, by the Abel–Jacobi identifications [10, Propositions 2.2 and 9.1]. Lemma 6.1 proves the first assertion.

For every $\sigma \in \mathrm{Aut}(\Omega/k)$, the morphisms f_Ω and σf_Ω have the same geometric outer homomorphism, by the G_k-compatibility of θ and invariance of geometric fundamental groups under algebraically closed extension. They are therefore equal. Their function-field map descends to k. One may see the descent directly from

$$\Omega(X)^{\mathrm{Aut}(\Omega/k)} = k(X).$$

To verify this equality, take an invariant fraction with a denominator $\sum_{i=1}^{r} a_i e_i$, where the $e_i \in k(X)$ are k-linearly independent, $a_1 = 1$, and r is minimal. Subtracting any conjugate of the denominator gives a denominator with fewer terms unless it is zero. All its coefficients, and then those of the numerator, are fixed and belong to k. The descended rational map extends by properness.

It remains to compare the arithmetic homomorphisms. We adapt the deck-transformation argument of [4, Lemma 1 and Corollary 2] to the common open geometric image. After geometric conjugation, assume f_* and θ agree on Δ_X, whose common image is open in Δ_Y. For $\gamma \in \Pi_X$, the element $z = f_*(\gamma)^{-1}\theta(\gamma) \in \Delta_Y$ centralizes $\theta(\Delta_X)$. If $z \neq 1$, choose open normal $N \triangleleft \Delta_Y$ with $N \subset \theta(\Delta_X)$ and $z \notin N$. On the corresponding proper hyperbolic cover $C \to Y_{\bar{k}}$, the nontrivial deck transformation defined by z induces the identity on $N^{\mathrm{ab}} = T_p J_C$. The faithfulness just proved makes this deck transformation the identity, a contradiction. Thus $f_* = \theta$ on the entire arithmetic group. □

Proposition 7.2. *The pro-p assertion of Theorem 1.2 holds when X and Y are smooth proper geometrically connected curves and Y is hyperbolic.*

Proof. Replace the target by the cover corresponding to the image of an open θ, so that θ is surjective. The induced homomorphism $T_p J_X \to T_p J_Y$ is surjective, so the differential a_θ of Proposition 6.4 is nonzero. Fix the resulting $f_\Omega : X_\Omega \to Y_\Omega$.

Let $V \to Y$ be an arithmetic Galois cover corresponding to an open normal subgroup of Π_Y, and let $W \to X$ be its θ-pullback. Both are geometrically connected over the same finite constant field E/K. The induced map $\Pi_W \to \Pi_V$ is again surjective. Proposition 6.4 gives $r : W_\Omega \to V_\Omega$. Naturality of its differential and Lemma 6.1 imply that r lies over f_Ω. They also imply equivariance under every geometric deck transformation. The resulting equivariant morphism $W_\Omega \to X_\Omega \times_{Y_\Omega} V_\Omega$ is a morphism between torsors under the same finite group, and is therefore an isomorphism. The geometric quotients so obtained are cofinal: every open subgroup of Δ_Y contains the intersection with Δ_Y of some open normal subgroup of Π_Y. Lemma 3.1(i) identifies the geometric outer homomorphism of f_Ω with that of θ. Lemma 7.1 now supplies the required K-morphism and equality of the full arithmetic homomorphisms. It also proves uniqueness.

A nonconstant morphism of proper curves is finite. Its function-field map has open Galois image, and the maps from the generic Galois groups to the respective fundamental groups are surjective [5, Exposé V, Proposition 8.2]. Hence its induced homomorphism is open. □

To allow an open source, it suffices to show that a homomorphism into the fundamental group of a proper hyperbolic curve kills the inertia at each missing point. The necessary local assertion is the following consequence of reconstruction for $\mathbf{G}_m$.

Lemma 7.3 (Vanishing of puncture inertia)**.** *For K p-subtame and Y/K proper hyperbolic, every G_K-homomorphism $\Pi_{\mathbf{G}_m} \to \Pi_Y$ kills $\Delta_{\mathbf{G}_m}$. Consequently every $\Pi_U \to \Pi_Y$, for a smooth separated curve U, factors through the smooth proper compactification of U, with unchanged image.*

Proof. Write $\Pi_{\mathbf{G}_m} = \mathbf{Z}_p(1) \rtimes G_K$, using the point 1. Suppose a generator has nontrivial image $z \in \Delta_Y$. Detect z in a finite arithmetic quotient and take the inverse image in Δ_Y of the cyclic subgroup generated by its image. This open subgroup is stable under the image section of G_K, since conjugation sends z to $z^{\chi_p(g)}$. It gives a proper hyperbolic cover $C \to Y$ to which the homomorphism lifts. The induced map $\mathbf{Z}_p(1) \to T_p J_C$ is nonzero, as is its differential by Proposition 4.2. Proposition 6.4 would produce a nonconstant $\mathbf{G}_{m,\Omega} \to C_\Omega$. Properness extends it to $\mathbf{P}^1_\Omega$, contradicting Riemann–Hurwitz.

For a boundary point of U, first make the point rational over a finite extension K'/K. A parameter and a compatible system of its roots give a splitting of the decomposition group of the henselian local field as $\widehat{\mathbf{Z}}(1) \rtimes G_{K'}$. The action on inertia is cyclotomic. Since the geometric kernel of Π_Y is pro-p, the resulting homomorphism factors through $\mathbf{Z}_p(1) \rtimes G_{K'}$. The first assertion kills this inertia. A finite cover extends étaleover the compactification precisely when these inertia groups act trivially: this is the normalization description of covers of a normal scheme [5, Exposé I, Corollaire 10.3]. Thus their closed normal closure is the kernel of the map to the fundamental group of the compactification. This proves the factorization; passage to the inverse limit over the open subcurves gives the same assertion for the generic point. □

Combining Proposition 7.2 with Lemma 7.3 proves the pro-p Hom theorem from every smooth separated curve to a proper hyperbolic curve. For a nonseparated smooth curve, apply this result on affine opens and glue by uniqueness.

8 Normal sources and open targets

We next remove the restrictions on the source and target. For the source, the essential point is to find a curve over a larger ground field which preserves every connected geometric cover of X. The curve theorem then gives a rational map on X. A separate extension argument shows that its compatibility with the fundamental group excludes indeterminacy, even when X is singular or nonseparated. For an open target, ramification at the missing points will similarly exclude intersection with the boundary.

8.1 A simultaneous curve reduction

An arbitrary curve in X need not preserve the image of θ on geometric fundamental groups. We instead use a generic pencil in the function field, with independent parameters in the ground field. Its regularity after every finite extension of the function field is what ensures the required surjectivity. Mochizuki obtains this surjectivity from a Bertini pencil and the Lefschetz theorem [12, Lemmas 16.1–16.2]; the following field-theoretic argument supplies the pencil without choosing a smooth projective model of X.

Lemma 8.1. *Let F/k be a regular finitely generated extension of positive transcendence degree d, in characteristic zero. There are algebraically independent variables $\mathbf{t}$ over F and a purely transcendental extension E/k contained in $F(\mathbf{t})$ such that $F(\mathbf{t})/E$ is regular of transcendence*

degree one. They can be chosen so that, for every finite k'/k and every finite extension M/Fk' regular over k', the extension $M(\mathbf{t})/Ek'$ is regular. If $F = k(X)$ for a normal geometrically integral finite-type scheme, there is a smooth geometrically connected curve U/E with function field $F(\mathbf{t})$ and a k-morphism $U \to X$ inducing a surjection $\Delta_{U/E} \to \Delta_{X/k}$.

Proof. We first prove the field-theoretic assertion underlying the construction. If M/k is regular and finitely generated and $a, b \in M$ are algebraically independent, then $M(t)/k(t, at+b)$ is regular. Let N_0 be the relative algebraic closure of $k(a, b)$ in M, which is finite over $k(a, b)$. For $z \in M(t)$ algebraic over $k(t, at + b)$, one has $z \in N_0(t)$. The intermediate field $k(a, b, t, z)$ is $L_0(t)$ for some $k(a, b) \subset L_0 \subset N_0$. Indeed, take a primitive element of $N_0/k(a, b)$. The coefficients of its minimal polynomial over $k(a, b, t, z)$ are algebraic over $k(a, b)$, hence belong to N_0; the subfield L_0 they generate has the asserted property by comparison of degrees. Thus $k(a, b, t, z)$ is stable under $\sigma : t \mapsto t+1$ fixing M. Put $u = at+b$ and $u' = u+a$. Then $k(a, b, t, z) = k(t, u, z)(u')$ is regular over $k(t, u')$, because $k(t, u, z)/k(t)$ is regular and u' is independent of it. But σz is algebraic over $k(t, u')$, so $\sigma z \in k(t, u')$, giving $z \in k(t, u)$.

Now choose a transcendence basis $x_1, \dots, x_d$ of F/k and put

$$u_i = x_i + t_i x_d \quad (1 \le i < d), \qquad E = k(t_i, u_i : 1 \le i < d).$$

The reversible change $x_i = u_i - t_i x_d$ gives $\operatorname{trdeg}_E F(\mathbf{t}) = 1$. Apply the pencil observation successively, also to every M/k' under consideration. It proves the asserted regularity with the same choices for all finite covers. Adding independent variables preserves their degrees and Galois groups.

It remains to realize this field construction geometrically. Choose a nonempty affine open of X. Clearing denominators of its finitely many coordinate functions in $F(\mathbf{t})$ realizes a nonempty smooth affine model U/E with a map to this open. Every connected geometric finite cover of X descends to some finite k'/k; its function field is one of the M above. The proved regularity keeps its pullback to $U_{\bar{E}}$ connected. This is exactly the asserted surjectivity on geometric fundamental groups, and hence on their pro-p quotients. □

The next lemma supplies the extension across the omitted locus of X. For smooth sources the extension is unconditional [12, Lemma 16.3], by the classical extension theorem for maps to abelian varieties. For a normal source we use the stated factorization on fundamental groups. The proof uses only abelian covers of the target. A positive-dimensional fibre in the closure of the graph would contain the whole target curve; the assumed factorization on fundamental groups would then trivialize covers of arbitrarily large degree over one fixed finite algebra.

Lemma 8.2 (Extension across a normal source)**.** *Let X/k be normal, geometrically integral and of finite type in characteristic zero, let $F = k(X)$, and let C/k be a smooth proper geometrically connected curve of genus $g \ge 1$. A rational map $q : \operatorname{Spec} F \to C$ extends uniquely to X if its geometric pro-p abelianization factors as*

$$\Delta_{\operatorname{Spec} F/k} \longrightarrow \Delta_{X/k} \longrightarrow T_p J_C.$$

Proof. Take the closure $\Gamma \subset X \times C$ of the graph. It is proper birational over X. Suppose it has a positive-dimensional geometric fibre over $\bar{x} = \operatorname{Spec} \Omega \to X$. Its reduced fibre is then the whole C_Ω. Choose an affine neighborhood $\operatorname{Spec} A$ of its generic point in Γ and let B be the normalization of A in F. This is finite since A is of finite type over a field. The algebra

$$T = B \otimes_A \Omega(C)$$

is nonzero and finite-dimensional over $\Omega(C)$.

Choose n with $p^{2gn} > \dim_{\Omega(C)} T$. After a finite constant extension k'/k, Lemma 3.2 realizes the quotient T_pJ_C/p^n as a geometrically connected abelian cover $V \to C_{k'}$ of degree p^{2gn}. The factorization assumption makes q^*V the generic fibre of a finite étaletorsor over $X_{k'}$: the kernel of $G_{Fk'} \to \Pi_{X_{k'}}$ is geometric and is killed by the assumed factorization. Over the normal scheme $\mathrm{Spec}(B \otimes_k k')$, its pullback and the pullback of V are isomorphic: finite étalecovers of a normal integral scheme are determined by their generic fibres [5, Exposé I, Corollaires 10.2–10.3]. A section of that torsor at $\bar{x}$, pulled back to T, therefore gives an $\Omega(C)$-algebra map $\Omega(V) \to T$. It is injective, since $\Omega(V)$ is a field. This forces $p^{2gn} \leq \dim_{\Omega(C)} T$, a contradiction.

Thus $\Gamma \to X$ is proper with finite fibres, hence finite [5, Exposé VIII, Théorème 6.1]. Being birational, it is an isomorphism by normality of X. This proves existence; separatedness of C proves uniqueness. □

Proposition 8.3. *The pro-p Hom theorem holds for normal geometrically integral finite-type sources and proper hyperbolic targets over p-subtame fields.*

Proof. In dimension zero both sides are empty. Suppose that $\dim X > 0$, let $\theta : \Pi_X \to \Pi_Y$ be open over G_K, and put $F = K(X)$. Apply Lemma 8.1 to obtain $U/E \to X$ with $E(U) = F(\mathbf{t})$. The field E is p-subtame by Lemma 2.1. The induced map

$$\Pi_{U/E} \longrightarrow \Pi_{Y/K} \times_{G_K} G_E = \Pi_{Y_E/E}$$

has the same geometric image as θ and is open. The curve case therefore gives $q \in Y(F(\mathbf{t}))$ inducing this homomorphism. For a proper curve of positive genus,

$$Y(F(\mathbf{t})) = Y(F) :$$

remove the variables one at a time, since a rational map $\mathbf{P}^1_F \dashrightarrow Y$ extends and is constant by Riemann–Hurwitz. The surjection $G_{F(\mathbf{t})} \to G_F$ shows that q_* agrees with θ on the entire generic arithmetic group. Lemma 8.2 extends q to X. Surjectivity of $G_F \to \pi_1(X)$ [5, Exposé V, Proposition 8.2] then gives the prescribed arithmetic homomorphism. Uniqueness follows by pulling two candidates back to U, applying the curve case, and descending equality from the generic point. □

8.2 Avoiding the boundary of the target

To treat an open target, we use proper compactifications of its finite covers. We recall two covering facts about a hyperbolic curve Y in characteristic zero: each geometric pro-p cusp inertia is a copy of $\mathbf{Z}_p(1)$, and Y has an arithmetic finite p-cover whose smooth compactified geometric components have genus at least two. Here "p-cover" refers to its geometric degree, not its constant-field degree. When Y is proper the second assertion is immediate. For an open Y, both assertions follow from the usual presentation of its geometric fundamental group [5, Exposé XIII, Corollaire 2.12]. If there are $r \geq 2$ boundary points, abelianizing this presentation gives residue vectors whose coordinates sum to zero. The vectors $(1, -1, 0, \ldots, 0)$ modulo p^a detect cusp inertia to arbitrary order. The cyclic cover with residues $(1, \ldots, 1, -(r-1))$ has, for $a > v_p(r-1)$,

$$2g(\bar{V}) - 2 = p^a(2g(\bar{Y}) - 2 + r) - (r-1) - p^{v_p(r-1)} > 0$$

when a is large. If $r = 1$, first take an unramified cover of degree p of $\bar{Y}$, which exists since $g(\bar{Y}) \geq 1$, and then apply the preceding construction. Finally descend the cover after finite constant extension and take an arithmetic Galois closure using Lemma 3.2.

The maps to these proper compactifications are supplied by Proposition 8.3. They descend to a map to $\bar{Y}$. We then use Mochizuki's ramification argument [12, proof of Corollary 14.2]: if that map met a missing point of Y, its order of vanishing there would have to be divisible by the ramification indices in every finite cover. The unboundedness of cusp inertia makes this impossible.

Theorem 8.4 (The pro-p Hom theorem)**.** *Theorem 1.2 holds for* $\Sigma = \{p\}$.

Proof. As before, replacing Y by the image cover reduces the proof to a surjective $\theta : \Pi_X \to \Pi_Y$. Choose an arithmetic Galois cover $V_0 \to Y$ as above, corresponding to $N_0 \lhd \Pi_Y$. For each open normal $N \subset N_0$, let $V \to Y$ be its cover, let $W \to X$ be the θ-pullback, and let E be their common constant field. Apply Proposition 8.3 to the induced open map from Π_W to the fundamental group of the compactification $\bar{V}$. It gives a unique dominant E-morphism $r_V : W \to \bar{V}$. Uniqueness gives compatibility with $V \to V_0$ and with all arithmetic deck transformations, including their semilinear action on E. Hence r_{V_0} descends to a dominant $\bar{f} : X \to \bar{Y}$.

Suppose $\bar{f}$ meets the boundary. Its inverse image is a nonempty effective Cartier divisor, so has a height-one point x. The local ring $\mathcal{O}_{X,x}$ is a DVR. If t is a parameter at the boundary point y, put $e = \operatorname{ord}_x(\bar{f}^* t) > 0$. For a cusp inertia generator z, choose $N \subset N_0$ with $z^e \notin N$. The order r of z in Π_Y/N does not divide e, and every branch of $\bar{V}$ above y has ramification index r. At a point $w \in W$ above x, the cover $W \to X$ is étale. A parameter s at $r_V(w)$ consequently gives

$$e = \operatorname{ord}_w(\bar{f}^* t) = r \operatorname{ord}_w(r_V^* s),$$

a contradiction. Thus $\bar{f}$ factors through $f : X \to Y$.

All r_V now land in V, and their arithmetic equivariance yields $W \simeq X \times_Y V$ as torsors, with identity on the constant quotients. Lemma 3.3 identifies f_* with the specified θ by one geometric conjugation. Uniqueness follows after pulling back to W_0 and using uniqueness for the proper target $\bar{V}_0$.

Finally, every dominant morphism $X \to Y$ induces an open homomorphism. The relative algebraic closure of $K(Y)$ in $K(X)$ is finite, and $K(X)$ is regular over it. Thus the generic Galois image is open. Surjectivity from generic Galois groups to fundamental groups of normal schemes [5, Exposé V, Proposition 8.2] proves the assertion for $\Pi_X \to \Pi_Y$. □

9 Passage to a set of primes

The passage from the pro-p theorem to a set of primes is due to Mochizuki. It appears for the full profinite statement in [12, Remark after Theorem 16.5], and for arbitrary Σ containing p in [13, Theorem A and the following remark, p. 121]. The same argument applies here because finite extensions of a p-subtame field are again p-subtame. One applies the pro-p theorem to each finite Σ-cover and uses uniqueness to recover the cover itself. We include the argument to make explicit its compatibility with the constant-field quotient.

Proof of Theorem 1.2. Let $\theta : \Pi_X^\Sigma \to \Pi_Y^\Sigma$ be open over G_K. Lemma 3.1 reduces it to a surjection. Passing to the canonical relative pro-p quotients gives a surjective $\theta_p : \Pi_X \to \Pi_Y$. Theorem 8.4 gives a unique dominant $f : X \to Y$ inducing θ_p.

For every open normal $N \lhd \Pi_Y^\Sigma$, let $V \to Y$ and $W \to X$ be the corresponding torsor and its θ-pullback. They have the same finite constant field E/K. By Lemma 3.1(ii), the restricted map is a surjection

$$\Pi_{W/E}^\Sigma \longrightarrow \Pi_{V/E}^\Sigma.$$

Pass to its own relative pro-p quotients and apply Theorem 8.4 over E, which is p-subtame. This gives a unique dominant E-morphism $r_V : W \to V$.

The composites $W \xrightarrow{r_V} V \to Y_E$ and $W \to X_E \xrightarrow{f_E} Y_E$ induce the same relative pro-p homomorphism over E. Pro-p uniqueness therefore makes them equal. For $a \in \Pi_Y^\Sigma/N$, the map $a_V^{-1} r_V a_W$ is E-linear, even if a_V and a_W are semilinear over E. It induces the same pro-p homomorphism as r_V, because the restricted homomorphism is equivariant under conjugation by a. The same uniqueness gives

$$r_V a_W = a_V r_V.$$

Thus the induced morphism is an isomorphism of (Π_Y^Σ/N)-torsors $W \simeq X \times_Y V$. Its action on the constant quotient is the identity, since r_V is over E. Lemma 3.3 now proves $f_* = \theta$ modulo one inner automorphism from Δ_Y^Σ.

Two morphisms inducing the same relative pro-Σ homomorphism modulo geometric conjugation induce the same such pro-p homomorphism, so are equal by Theorem 8.4. Openness for dominant morphisms is the same generic-Galois argument used there. This proves (1.2) for every Σ containing p. □

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